Hand-to-Mouth Agent Classification for Brazil

Kaplan–Violante–Weidner (2014) applied to POF 2017–18 & PNADC

H2M Classification

Goal: Classify Brazilian households into three agent types (Kaplan, Violante & Weidner 2014)

- **PH2M** (Poor HtM) — low liquid, low illiquid assets
- **WH2M** (Wealthy HtM) — low liquid, significant illiquid assets
- **Ricardian** — sufficient liquid buffer to smooth consumption

Type shares estimated in POF $\rightarrow$ transferred to PNADC via bin-matching & Monte Carlo $\Rightarrow$ state$\times$quarter panel for HANK. **Literature targets**

(Brazil): Carvalho & Zilberman (2022); De Souza (2023) — PH2M 20–28%, WH2M 12–18%, Ricardian 55–65%, Total HtM 35–45%.

Data Sources

Survey	Period	Sample	Format	Key variables
POF 2017–18	Jul 2017–Jul 2018	$\sim$140k (age $\geq$ 15)	Fixed-width (5 tbl)	Income, assets (imputed), expenditure, weights
PNADC	2017 Q1–2018 Q4 (8 qtrs)	varies	CSV (test5/6/7)	Demographics, income, employment, weights

POF Table Structure

POF: five fixed-width text files (column positions from Excel data dictionary)

File	Key columns
`DOMICILIO.txt`	COD_UPA, NUM_DOM, UF, PESO_FINAL
`MORADOR.txt`	V0403 (age), V0404 (sex), NIVEL_INSTRUCAO, RENDA_TOTAL
`RENDIMENTO_TRABALHO.txt`	V8500_DEFLA (deflated labour inc.), V5302/V5303 (formality)
`OUTROS_RENDIMENTOS.txt`	QUADRO 55=pension, 56=govt transfers, 57=financial income
`ALUGUEL_ESTIMADO.txt`	V8000_DEFLA (estimated rent $\rightarrow$ real estate proxy)

Pipeline Architecture

1. POF: Load & merge 5 tables $\rightarrow$ classify each person
2. POF: 6D bins $\rightarrow$ weighted shares (Dirichlet $\alpha = 0.1$)
3. PNADC: Same 6D bins $\rightarrow$ left-merge POF shares
4. PNADC: Monte Carlo type draw (seed=42)
5. PNADC: Aggregate to state$\times$quarter shares
6. PNADC: Generate choropleth maps per quarter

Zero-Financial-Income Problem

Only **8%** of POF individuals report financial income (QUADRO 57). Naive: liquid $=$ (fin. inc. $\times$ 12)/SELIC $\Rightarrow$ 92% get zero, all HtM. Grid

search (864 combinations): best achievable total HtM $\approx$ 91.5% vs target $\approx$ 40%.

Augmented Liquid Asset Imputation

$$\text{liquid} = \underbrace{\frac{\text{fin. inc.} \times 12}{\text{SELIC}}}_{\text{fin.}} + \underbrace{\text{pen.} \times 1}_{\text{buf.}} + \underbrace{\max(0,\ \text{R_T} - 12 \times \text{inc.}) \times 0.5}_{\text{savings}}$$

Savings proxy zeroed for Bolsa Família recipients (QUADRO 56 > 0).

Two-Threshold Classification Rule

$$\text{type} = \begin{cases} \text{Ricardian} & \text{liquid} > \lambda \\ \text{WH2M} & \text{liquid} \leq \lambda,\ \text{illiquid} \geq \mu \\ \text{PH2M} & \text{else} \end{cases}$$

$\lambda = 0.50$, $\mu = 3$ (monthly income).

Calibrated Parameters

Parameter	Value	Description
SELIC_RATE	0.09	Annual rate for capitalising financial income
LIQUID_THRESH	0.50	Liquid assets / monthly income threshold (λ)
ILLIQUID_MULT	3	Illiquid assets / monthly income threshold (μ)
PENSION_MULT	1	Months of pension held as liquid buffer
SAVINGS_FRAC	0.50	Fraction of income surplus treated as savings
POVERTY_LINE	R$170	BRL/month per capita (Bolsa Família line, diagnostic)

Six Bin Dimensions

Dimension	Values	Notes
macro_region	North, NE, SE, South, Central_West	UF → region cut
age_group	15–24, ..., 65+ (6-way emp.; 3-way inact.)	Coarser for inactive
gender	male, female	POF V0404; PNADC V2007
education_group	no_ed., primary, secondary, tertiary	POF NIVEL_INSTRUCAO; PNADC VD3004
inc_band	B1–B5 (BRL [170, 700, 2k, 4k]); no_income inact.	Absolute bands, same in both
labour_3way	employed, unemployed, inactive	formal+inf+SE → employed

Mismatch Problem

Initial 6D bin scheme: **60.2% match** (355k of 893k PNADC individuals unmatched). **Root causes:**

1. Labour-status label mismatch: POF 3-way vs PNADC 5-way (formal/informal/SE/unemp/inactive). Dropping labour alone $\rightarrow$ 99.8% match.
2. Relative quintiles: Q1 in POF $\neq$ Q1 in PNADC; 100% of Q1$\times$informal unmatched.

Illustrative; actual values depend on PNADC panels loaded.

Mismatch Fix — Before/After

Dimension	Before	After
labour_status	5-way (formal/informal/SE/unemp/inact.)	3-way (formal+inf+SE → employed)
income_band	Relative per-survey quintile (Q1–Q5)	Absolute BRL bands [170, 700, 2k, 4k]
inactive income	Quintile assigned normally	Fixed `no_income`
age_group (inactive)	6-way for all	3-way (15–34, 35–54, 55+)

Strategy Comparison

Strategy	POF bins	Match rate	Trade-off
A: Original (relative quintile + 5-way labour)	3,102	60.2%	Baseline — too many unmatched
D: Drop labour entirely (5D)	1,180	99.8%	Loses employment heterogeneity
G: 3-way labour + conditional absolute bands	1,361	∼77%	Best balance; chosen
I: Drop labour, conditional absolute bands	1,361	99.9%	Near-perfect match, no labour dim.

Strategy G: 1,361 bins; only 1 small bin (<30 weighted obs).

National Validation

Stage	PH2M	WH2M	Ricardian	Total HtM
1. POF Classification	21.5%	19.3%	59.2%	40.8%
3. PNADC Post-Merge	25.0%	28.0%	47.0%	53.0%
4. PNADC Post-MC	24.9%	28.0%	47.0%	53.0%
Literature target	20–28%	12–18%	55–65%	35–45%

Monte Carlo realised shares track expected within $\sim$0.1 pp.

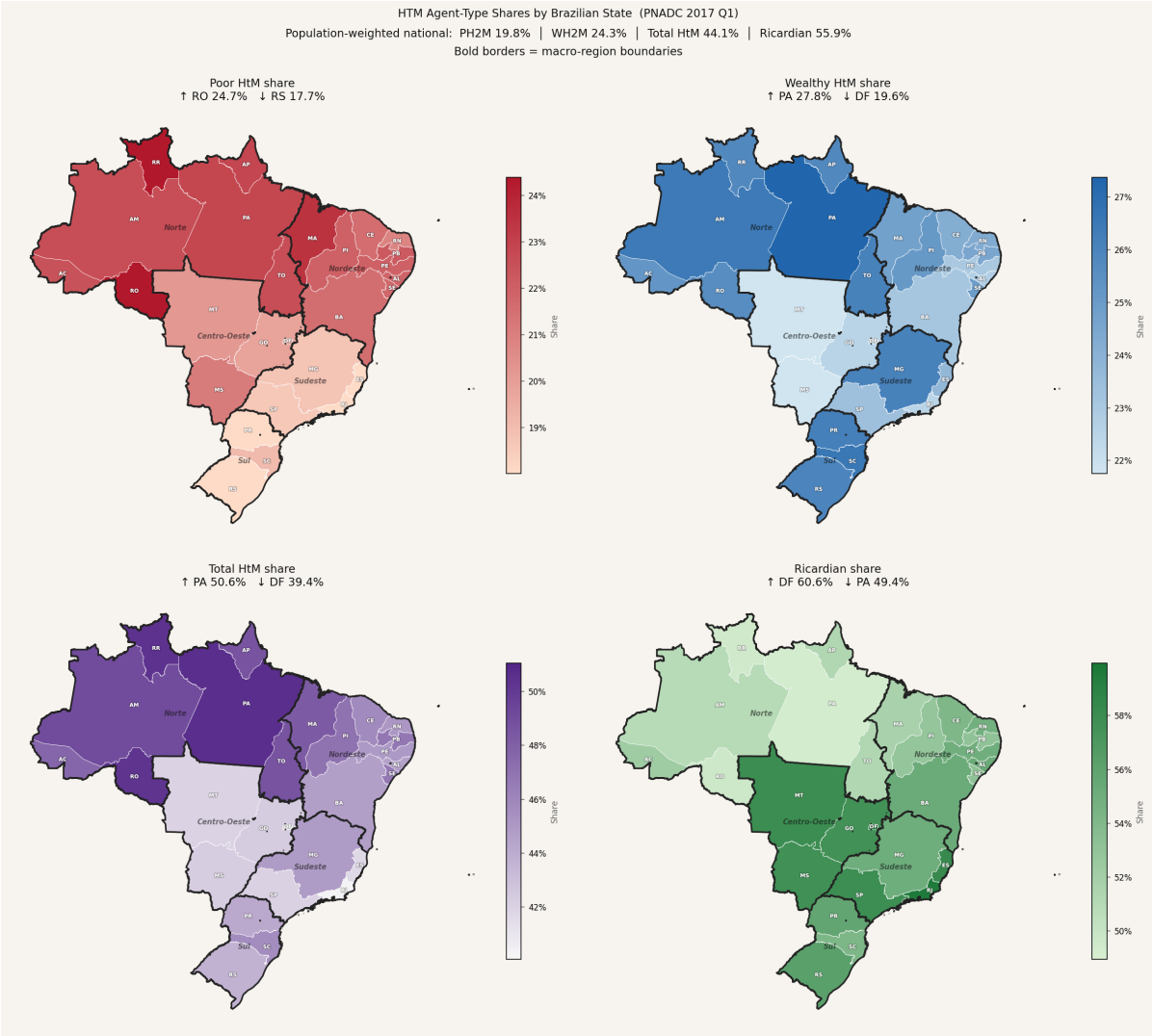
Warning

PNADC WH2M (28%) elevated vs POF (19.3%) and literature (12–18%). Driven by 22.7% unmatched rows falling back to national averages, and inactive individuals matched to high-illiquid POF bins.

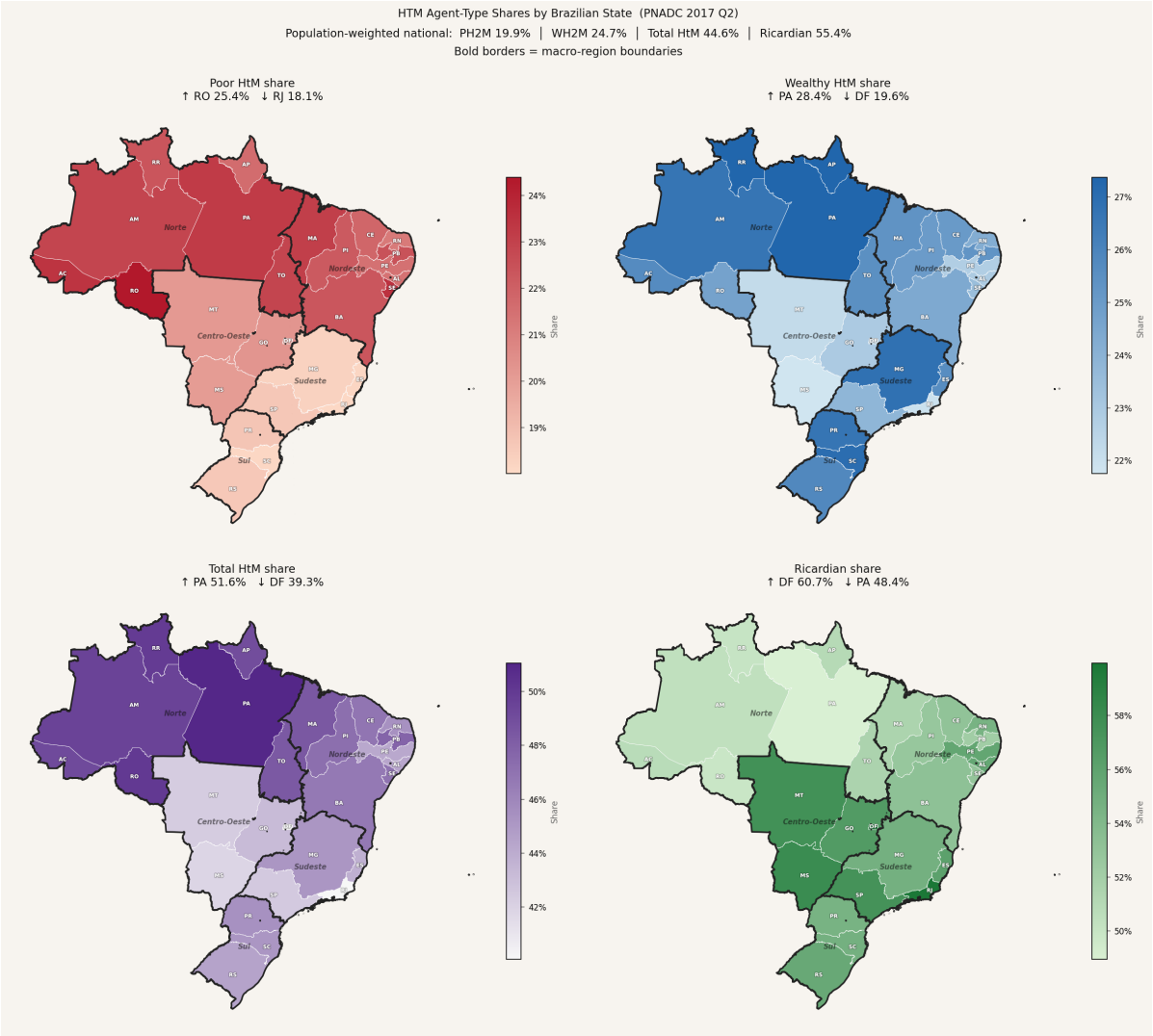
Key Regional Patterns

- **PH2M** highest in **Northeast** (e.g. Maranhão 26.4%) — poverty, low financial inclusion
- **WH2M** highest in **North** (e.g. Amazonas 22.4%) — land/housing but low liquid access
- **Ricardian** highest in **Southeast/South** (e.g. São Paulo 64.5%) — financial development, formal labour
- **Total HtM**: clear North/Northeast vs South/Southeast gradient — heterogeneity for HANK

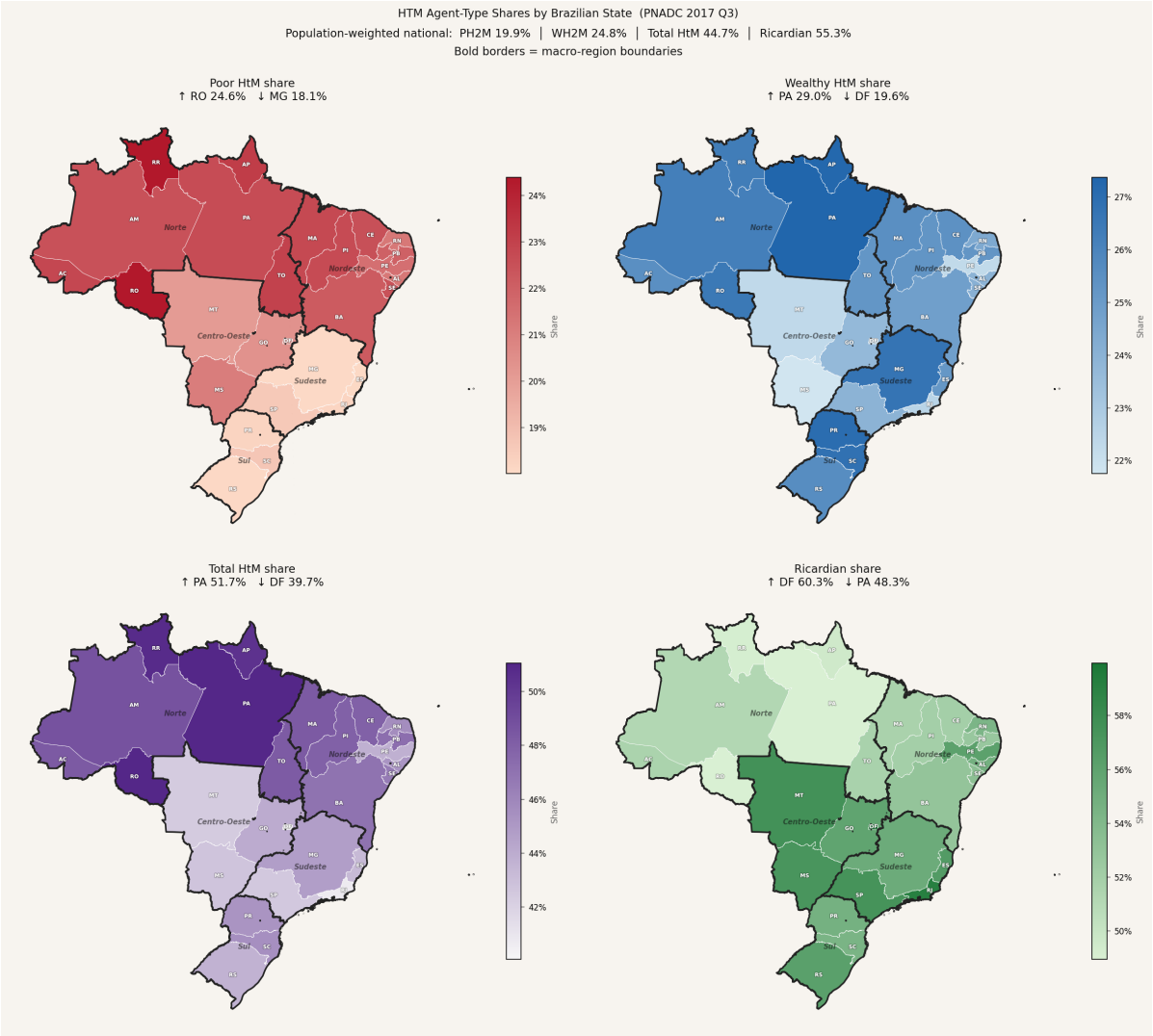
Choropleth Maps — 2017 Q1



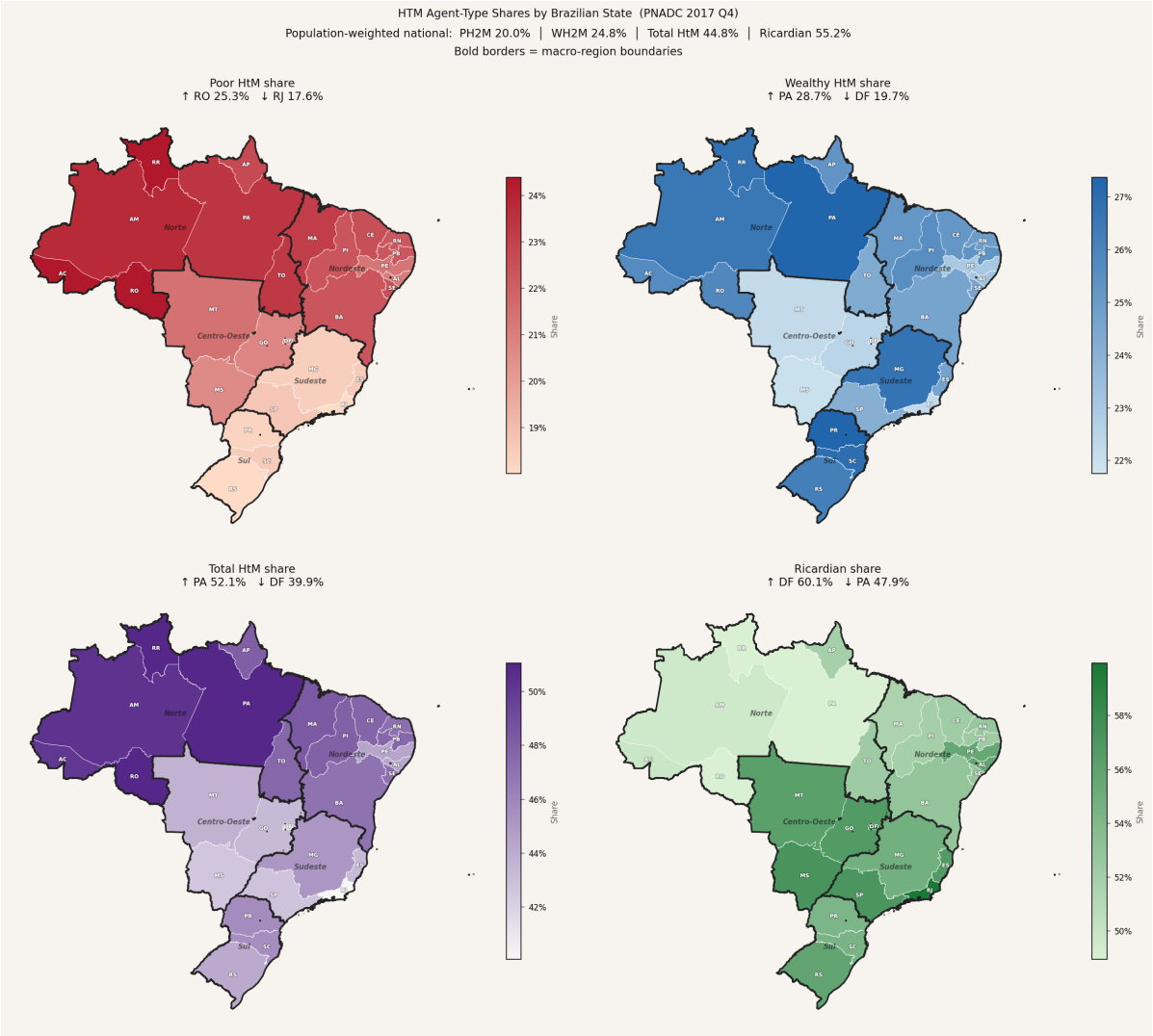
Choropleth Maps — 2017 Q2



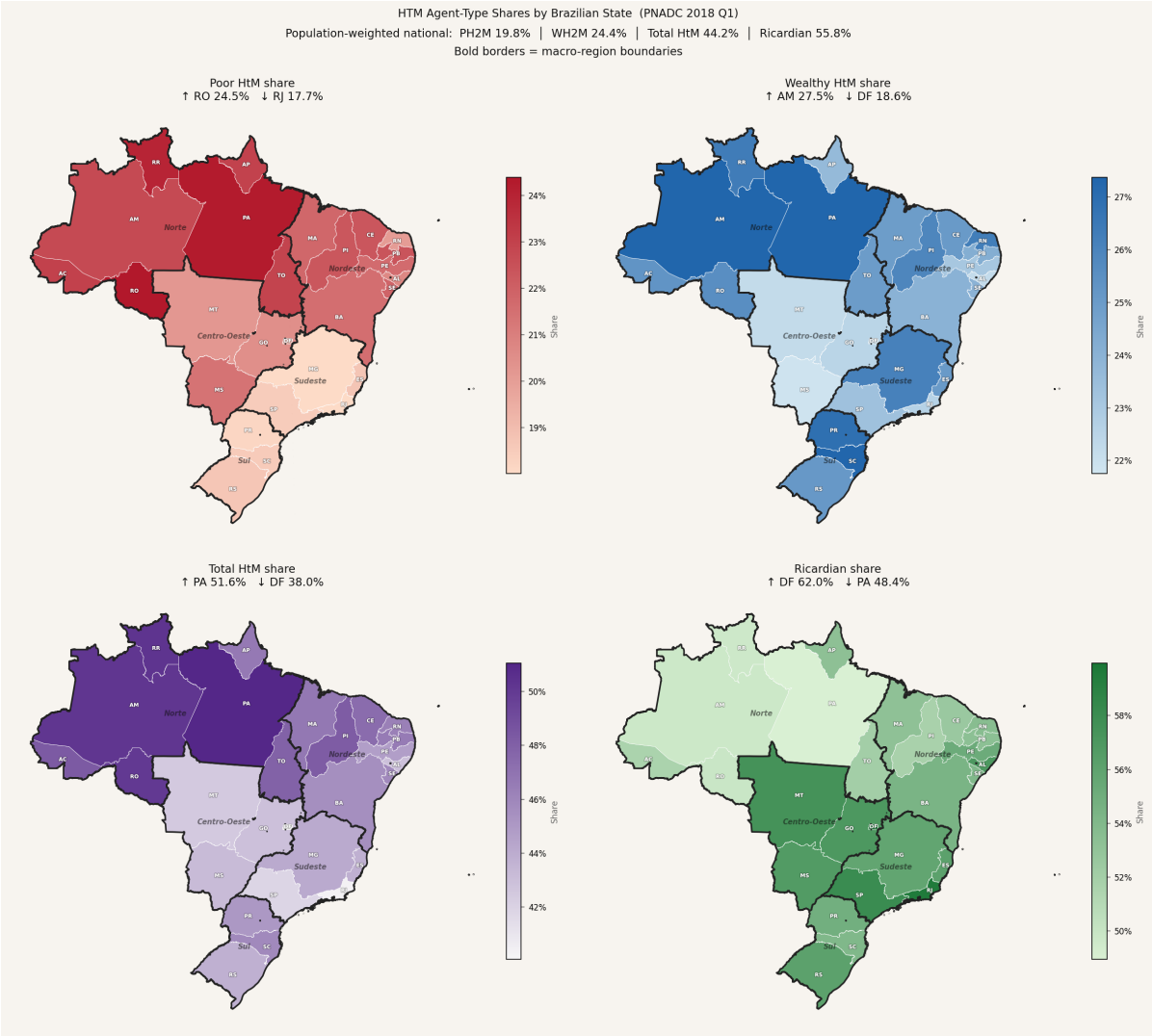
Choropleth Maps — 2017 Q3



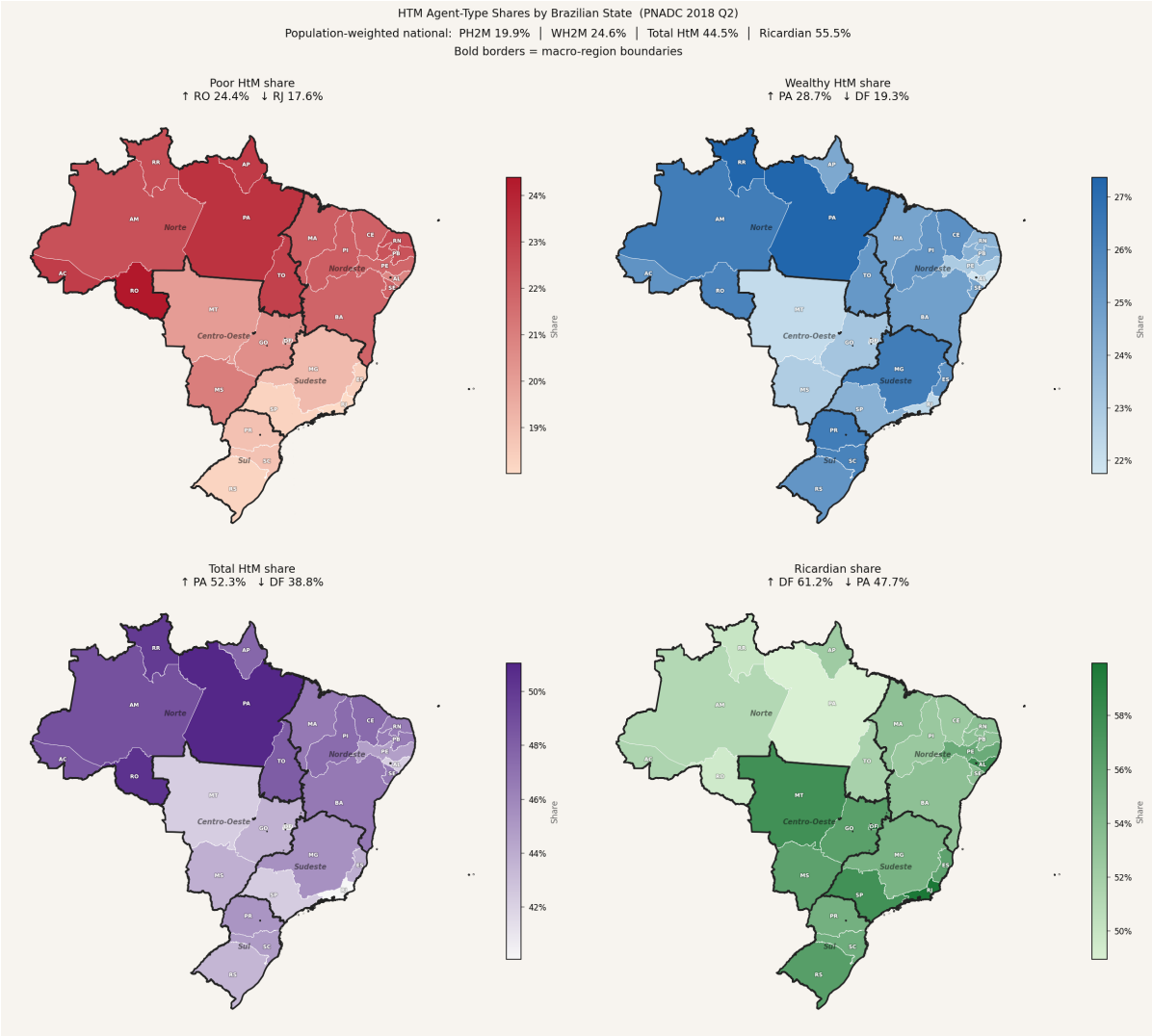
Choropleth Maps — 2017 Q4



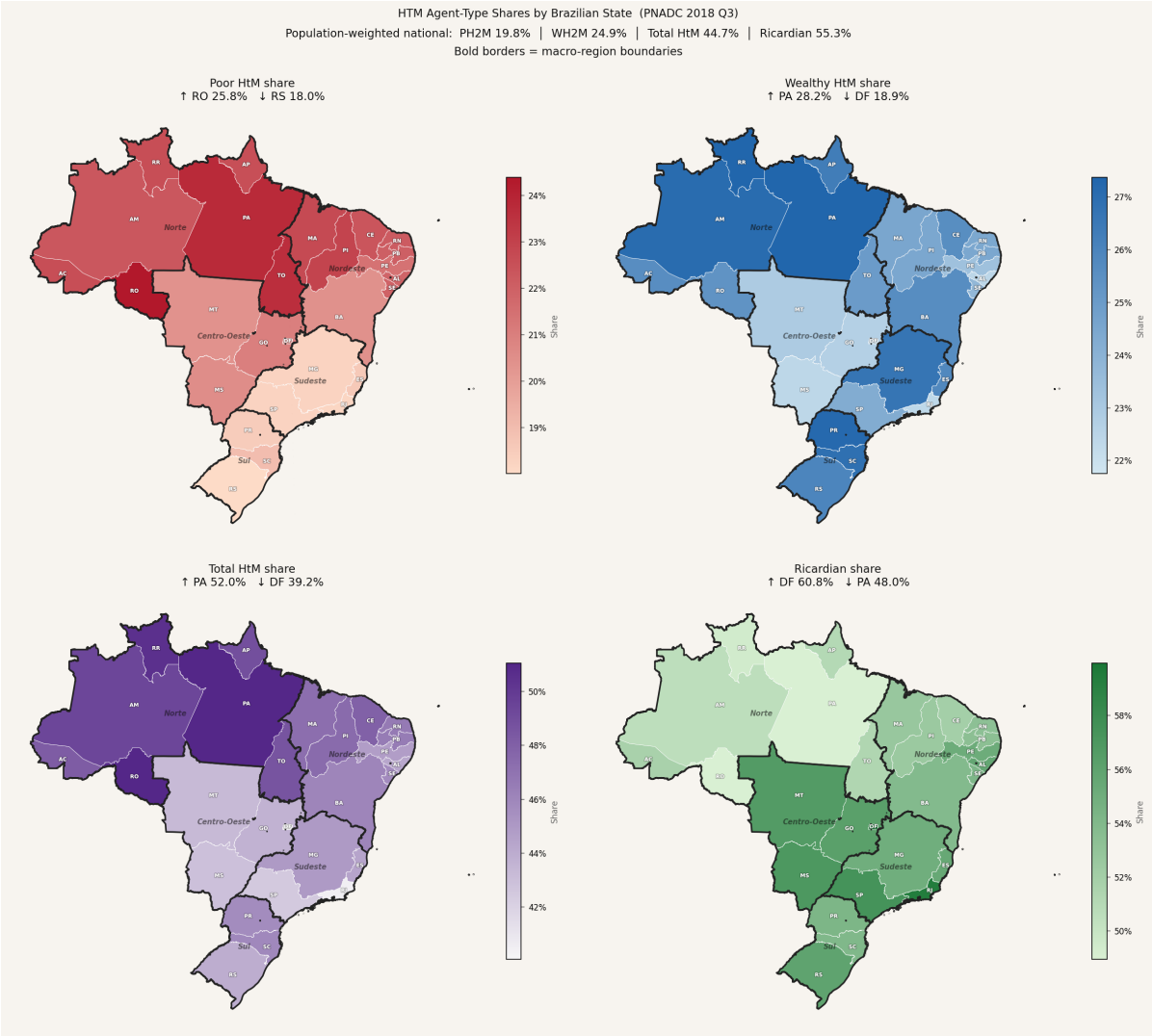
Choropleth Maps — 2018 Q1



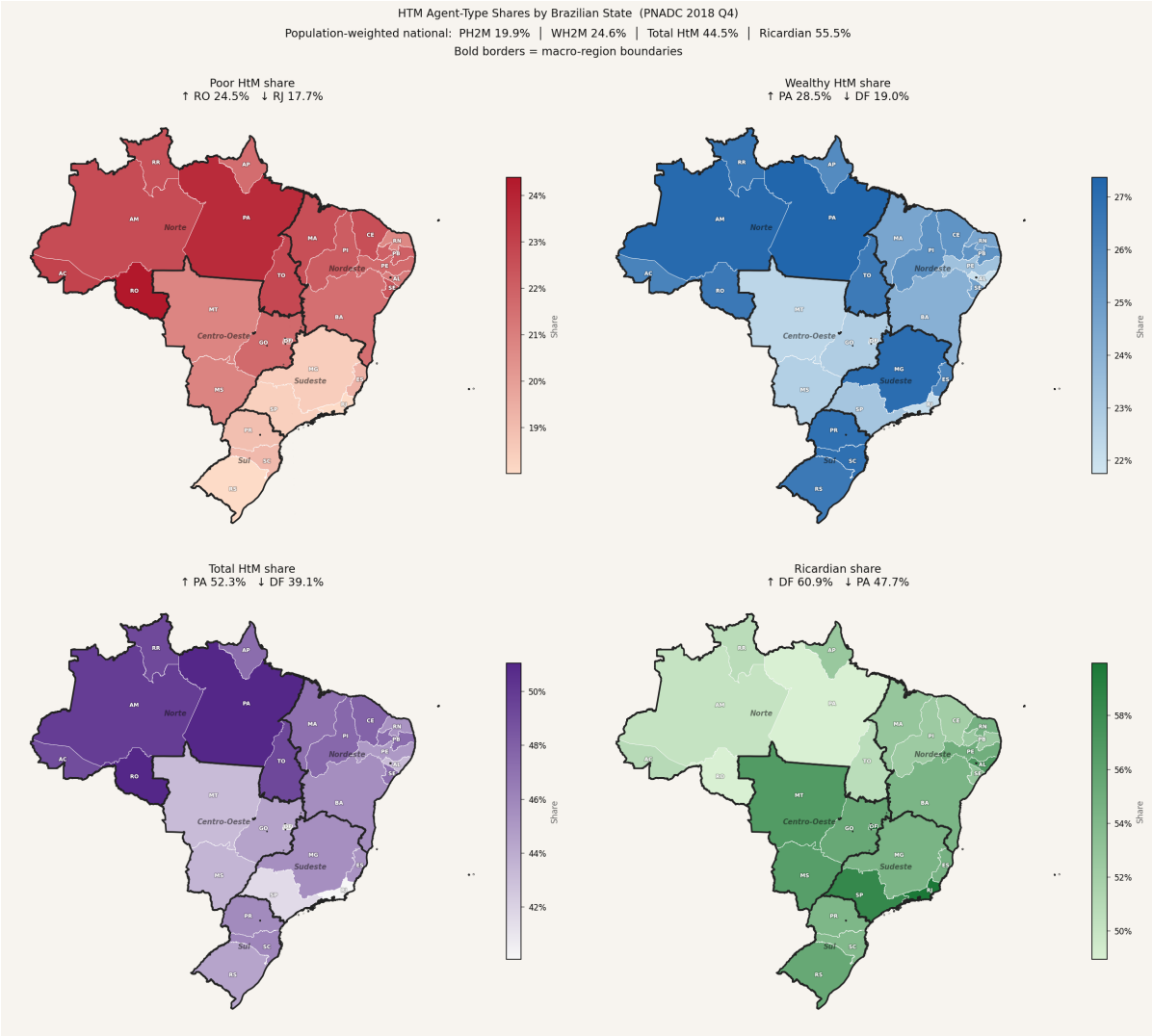
Choropleth Maps — 2018 Q2



Choropleth Maps — 2018 Q3



Choropleth Maps — 2018 Q4



Robustness

λ swept 0.25–1.50 in steps of 0.25; other params fixed. **Findings:**

- PH2M highly sensitive: $\lambda - 0.25 \Rightarrow -10$ pp; $\lambda + 0.25 \Rightarrow +18$ pp
- WH2M stable: $\pm$3–4 pp; determined by illiquid threshold
- $\lambda = 0.50$: only threshold where all four types sit within literature bands

Output Files

- `pof_bin_shares.csv` — demographic bin shares (1,361 bins), Dirichlet smoothing
- `state_quarter_htm_shares.csv` — $27 \times 8 = 216$ rows (weighted)
- `choropleth_htm_YYYYQN.png` — one 4-panel map per quarter (PH2M, WH2M, total HtM, Ricardian)